# Days 05 & 06 — Installing & Configuring Monitoring Tools

## Exercise Pack: Step-by-step Demo + Hands-on Lab 3

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**Course:** Preventive Surveillance in Cybersecurity — 420-D01-AB **Cohort:** CYB-01 **Module:** Installing and Configuring Monitoring Tools (Days 05 & 06) **Total contact hours:** 4 h hands-on (2 h Demo + 2 h Lab 3) **Author / Sole IP owner:** Cyber.SoHo Educational Hub **Tagline:** *From the Industry to the Classroom — Building the IT's Future, Today and Together!*

**Hard deadline for both deliverables in this pack: Saturday, 31 May, 23:59 EST**, uploaded to the OmniVox LEA (Léa, the Omnivox student portal section that hosts course material and DropBoxes) **DropBox** of course **420-D01-AB**.

**File-name convention:**

- Day 05 Demo write-up: `CYB01_<surname>_<firstname>_Lab01_LabEnvironment.pdf`
- Day 06 Lab 3 write-up: `CYB01_<surname>_<firstname>_Lab03_MonitoringStack.pdf`

If you upload after the cut-off the OmniVox LEA system stamps your submission as late automatically — the grader does not have to decide. Aim to upload at least one hour before the deadline; you do not want to discover at 23:55 that the campus Wi-Fi is having a bad evening.

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## How this pack fits into the bigger picture

Your **Course Procedure Manual** is the cumulative deliverable of 420-D01-AB. Over the semester you will hand in **six Labs**, each producing one chapter:

| # | Lab | Day(s) | Chapter |
|---|---|---|---|
| 1 | Lab Environment Setup | 05 (this pack) | Chapter 1 |
| 2 | (covered in earlier sessions — already submitted) | — | already submitted |
| 3 | Monitoring Stack (Wireshark / SNMP / Wazuh / Splunk) | 06 (this pack) | Chapter 2 |
| 4 | Vulnerability Assessment | 08 | Chapter 3 |
| 5 | Hardening & Baselines | 10 | Chapter 4 |

| # | Lab | Day(s) | Chapter |
|---|---|---|---|
| 6 | Incident Response & Forensics walk-through | 12 | Chapter 5 |

The final, bound **Course Procedure Manual** is the assembly of all six chapters into a single PDF, due on **Class 14**. The two chapters in this pack count for **10 %** of the final grade (5 % each).

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# The reporting standard — please read before you start

We do not grade your ability to follow a recipe. We grade **whether your written chapter could teach this material to someone who has never heard of Wireshark, Wazuh or Splunk**. A neighbour who is mildly curious, a junior colleague, your future self in six months.

Concretely, this means:

- **Every step is numbered**, named, and explained in your own words — never *"click here, then click there"*. State **what** you are doing, **why** it works, and **what** you expect to see.
- **Every screenshot has a caption** that names the figure (Figure 7, Figure 8…) and one sentence of context. A screenshot without a caption is decoration; a captioned screenshot is evidence.
- **Every command is shown verbatim** in a code block, followed by a short explanation of any flag a beginner would not recognise.
- **Every command output that proves the step worked** is captured as a screenshot or a fenced text block — *"trust me, it ran"* is not acceptable.
- **Every link is verified** with VirusTotal URL scanner before you paste it, and you write the URL in full so the reader does not have to hover.
- **Every troubleshooting incident** you actually hit during the lab goes into a *Troubleshooting* section: the symptom, the diagnosis, the fix, with citations.
- **The Reflection section is 200 words minimum**, in your voice, answering: what surprised me, what I would do differently next time, what I would tell a future student to watch out for.

If at any point you are unsure whether your chapter is detailed enough, ask yourself: *could a determined stranger sit down with my PDF and zero prior knowledge, and reach the same end state?* If the answer is no, the chapter is not finished.

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# Day 05 — Step-by-step Demo (2 hours)

## Building your Lab Environment on VirtualBox, GNS3 and AWS Academy Learner Lab

### Chapter 1 of your Course Procedure Manual

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## Why this exercise matters

Every Lab in the rest of this course assumes you can spin up a clean Windows or Linux VM (Virtual Machine) in under five minutes, snapshot it, network it to a router VM, and tear the whole thing down without leaving wreckage on your laptop. If you cannot do that fluently, every future Lab takes triple the time and you spend the semester fighting your environment instead of learning the material.

This is also the **first chapter you will hand in** under the new reporting standard. Your text must be good enough for a complete stranger to recreate the exact same lab from your document alone.

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## What you will build

```
                          AWS Academy Learner Lab (cloud)
                          +-----------------------------+
                          |  Ubuntu EC2  (CloudWatch)  |
                          +-------------+--------------+
                                        |  (optional VPN tunnel later)
                                        |
   Your laptop (host) ------------------+
   +--------------------------------------------------+
   |  VirtualBox host-only network 10.10.10.0/24      |
   |  +---------+   +---------+   +----------------+   |
   |  | Win10   |   | Ubuntu |   | Kali Linux    |   |
   |  | victim |   | server |   | analyst tool  |   |
   |  +---------+   +---------+   +----------------+   |
   |                |              |                     |
   |  +-----------------------------+                  |
   |  |  OPNsense FW (GNS3-linked) |                  |
   |  +-----------------------------+                  |
```

```
|  GNS3 topology bridged to VirtualBox          |
+-----------------------------------------------+
```

The three environments serve different purposes:

| Environment | What it gives you | What it cannot do |
|---|---|---|
| VirtualBox on your laptop | Full Windows / Linux VMs, snapshots, free, offline | Cannot easily simulate a multi-router enterprise topology |
| GNS3 (Graphical Network Simulator-3) | Real router / firewall images, layer-2/3 realism, OPNsense / pfSense, Cisco IOS-on-Linux | RAM-hungry; you need separate VMs for endpoints |
| **AWS Academy Learner Lab** | Real cloud — CloudWatch, GuardDuty, real public IPs | Time-limited credits; you must clean up after each session |

A working analyst rarely sits in only one of these. Your résumé is stronger if you can demonstrate competence in all three.

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# Phase 0 — Pre-flight checklist (10 min)

Tick every item **before** installing anything. Take a screenshot of your laptop's *About / System info* panel and include it as **Figure 1** in your chapter.

- [ ] CPU supports virtualisation (Intel VT-x or AMD-V) — verify with `systeminfo | findstr Hyper-V-Requirements` on Windows, or `lscpu | grep Virtualization` on Linux.
- [ ] At least **8 GB RAM** (16 GB recommended).
- [ ] At least **60 GB free disk** for the next few Labs.
- [ ] Virtualisation **enabled in BIOS** (Basic Input / Output System) **or UEFI** (Unified Extensible Firmware Interface) — Intel VT-x or AMD-V toggle.
- [ ] On Windows 10/11: **Hyper-V disabled** via *Turn Windows features on or off* — VirtualBox and Hyper-V do not get along.
- [ ] An AWS Academy invitation already in your student mailbox (the program coordinator sends it on enrolment).

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# Phase 1 — Install VirtualBox (15 min)

**Reference:** VirtualBox downloads.

1. Open the Downloads page in a browser. Pick the installer for your host OS (Operating System).
2. Run the installer with the default selections. It will briefly disconnect your network adapter while it installs the bridged / host-only drivers — that is normal and expected.
3. Install the matching **Extension Pack** (same version number) from the same page. The Extension Pack adds USB (Universal Serial Bus) 2.0 / 3.0 pass-through, an RDP (Remote Desktop Protocol) server, PXE (Preboot Execution Environment) boot — useful in later Labs.
4. Launch *VirtualBox Manager* once to confirm it starts cleanly.

**Screenshot — Figure 2:** *VirtualBox Manager window showing version number in the title bar.*

📺 **Video (≈ 10 min):** Search the The PC Security Channel for *"Install VirtualBox 7"*. Watch from the official channel, do not download or rehost.

## Create a host-only network

1. In VirtualBox Manager open *File → Tools → Network Manager → Host-only Networks → Create*.
2. Name it `vboxnet-cyb01`, set IPv4 to `10.10.10.1/24`, **disable** DHCP (Dynamic Host Configuration Protocol) — we want predictable IP addresses.
3. Save.

**Screenshot — Figure 3:** *Network Manager dialog with the new `vboxnet-cyb01` highlighted and the IPv4 / DHCP-off settings visible.*

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# Phase 2 — Download and prepare the VM images (20 min)

For every download below: verify the SHA-256 hash against the vendor page, **and** paste the download URL into VirusTotal before clicking it.

| VM | Official source |
|---|---|
| **Windows 10 / 11 Evaluation** (90-day trial) | Microsoft Evaluation Center |
| **Ubuntu Server 22.04 LTS** | ubuntu.com/download/server |
| **Kali Linux** prebuilt VirtualBox OVA (Open Virtualisation Appliance) | kali.org/get-kali |

| VM | Official source |
| --- | --- |
| **OPNsense** ISO (the firewall we will run) | [opnsense.org/download](opnsense.org/download) |

For each downloaded image:

1. Verify SHA-256 — on Linux / macOS `shasum -a 256 file.iso`, on Windows `Get-FileHash file.iso`.
2. Compare the result to the hash on the vendor page.

**Screenshots — Figures 4 to 7:** *each verification result alongside the vendor's posted hash, both visible on screen so the grader can see they match.*

### Create the four VMs

For every VM:

1. *VirtualBox Manager → New → Name =* `cyb01-<role>` *→ Type / version matching the ISO.*
2. Allocate RAM: Ubuntu = 2 GB · Windows = 4 GB · Kali = 4 GB · OPNsense = 1 GB.
3. Disk size: 25 GB for Ubuntu and OPNsense · 60 GB for Windows · 40 GB for Kali.
4. **Adapter 1 = Host-only → `vboxnet-cyb01`. Adapter 2 (Ubuntu / Windows / Kali only) = NAT** (Network Address Translation) for outbound package downloads.
5. For OPNsense: **Adapter 1 = NAT (WAN — Wide Area Network)**, **Adapter 2 = Host-only (LAN — Local Area Network)**.
6. Attach the ISO, start the VM, install the OS. Use consistent classroom credentials `cyber / Cyber.SoHo2026` so screenshots are reproducible — **change them in production**.
7. After each install, *VM → Snapshots → Take Snapshot → name* `00-fresh-install`.

**Screenshots — Figures 8 to 11:** *running desktop / login banner of each VM with the IP address visible on screen.*

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# Phase 3 — Install GNS3 (20 min)

**Reference:** [GNS3 download](GNS3 download).

1. Create a free GNS3 account (required to download).
2. Download the *all-in-one* installer for your host OS.
3. During install, accept the **GNS3 VM** option — it ships as an OVA.
4. Import the GNS3 VM into VirtualBox: *File → Import Appliance → select the GNS3 OVA file.*
5. Start the GNS3 VM. Note its host-only adapter IP address.
6. In the GNS3 desktop app: *Edit → Preferences → GNS3 VM → enable*. Point it at the running VM IP.

7. *Edit → Preferences → VirtualBox VMs → New → pick* `cyb01-opnsense`. This step lets you drag your OPNsense VM into a GNS3 topology.
8. Build a tiny topology to verify the wiring: drag an Ethernet switch + the OPNsense VM + a generic VPCS (Virtual PC Simulator) host onto the canvas. Connect them. Start. Verify ping reaches from the VPCS to the OPNsense LAN.

**Screenshot — Figure 12:** *running GNS3 topology with green link indicators between all nodes.*

📺 **Video (≈ 14 min):** Search the [David Bombal channel]() for *"GNS3 install with VirtualBox"*. Open the channel directly, do not download the video.

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# Phase 4 — AWS Academy Learner Lab access (15 min)

1. Open the invitation email from AWS Academy. Click *Accept invitation*.
2. Create an account on [awsacademy.instructure.com]() using the same email address as the invitation.
3. Open the course → *Modules* → *Learner Lab*.
4. Click *Start Lab* and wait for the small *AWS* icon to turn green.
5. Click *AWS* — a new browser tab opens directly in the AWS Management Console with temporary credentials already injected.
6. Launch your first **t2.micro Ubuntu EC2** (Elastic Compute Cloud) instance — region `us-east-1`, default VPC (Virtual Private Cloud), allocate an Elastic IP if you want a stable address, security group with SSH (Secure Shell) from your *home IP only*.
7. Generate a key pair, download the `.pem` (Privacy Enhanced Mail) file, set permissions: on Linux / macOS `chmod 400 mykey.pem`.
8. SSH in: `ssh -i mykey.pem ubuntu@<elastic-ip>`.

**Screenshots — Figures 13 & 14:** *AWS Management Console with the running EC2 instance + a working SSH session in your local terminal.*

⚠️ **Always click *End Lab* when you finish.** Academy credits are limited per cohort and they drain even when you are not actively using the lab. Leaving the lab running overnight is the single most common way students lose access mid-semester.

📺 **Video (≈ 12 min):** Search YouTube for *"AWS Academy Learner Lab walkthrough"*. The official AWS Educate channel publishes a current one; use the link, do not download or mirror the video.

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# Phase 5 — Verification (20 min)

Run each verification from the host indicated. Capture a screenshot for every command, even if the output looks identical to the last — the grader needs to see that you actually ran them all.

From the **Kali VM**:

1. `ping 10.10.10.1` (OPNsense LAN) — should succeed.
2. `nmap -sn 10.10.10.0/24` — list of live hosts.

From the **Ubuntu VM**:

3. `ping <kali-ip>` — should succeed.

From the **Windows VM**:

4. Open a browser → `https://10.10.10.1/` → the OPNsense login page should load (browser will warn about the self-signed certificate; that is expected).

From **your laptop host**:

5. SSH again to the AWS EC2 instance — should succeed.

**Screenshots — Figures 15 to 19:** *one per verification command, each captioned with what was verified.*

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# Phase 6 — Snapshot everything (5 min)

For every VM, take a fresh snapshot: *VM → Snapshots → Take Snapshot → name `01-networked`*. You will roll back to this point at the start of every future Lab. This is your save point.

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# Phase 7 — Write your manual chapter (homework, ≈ 90 min)

Use the Course Procedure Manual chapter template (provided on the LEA portal as `Chapter_Template.md`; you may format in Word, Pages, LibreOffice or Markdown-to-PDF — the only requirement is final PDF). Required sections, in this order:

1. **Title page** — course code, your name, student number, date, chapter number, version.
2. **Purpose** — five lines maximum, in your own words.
3. **Lab topology diagram** — redraw the ASCII diagram above in draw.io or by hand on paper and photograph it. Do not reuse the diagram from this hand-out verbatim.
4. **Procedure** — every phase above, numbered, every figure captioned with at least one sentence of context.
5. **Verification table** — columns: IP address, source, target, command, result, screenshot reference. Every row corresponds to one verification you actually performed.
6. **Troubleshooting** — at least three real problems you hit and how you solved them. Examples to inspire you: *"VirtualBox refused to start the VM with VERR_VMX_NO_VMX — solved by enabling VT-x in BIOS"*; *"GNS3 VM stuck at boot — fixed by re-importing with nested virtualisation disabled"*.
7. **Reflection** — minimum 200 words in your own voice: what surprised you, what you would change next time, which environment you trust most for which job, what you would tell a future student to watch out for.
8. **Sources** — every link you used, every video, every documentation page. Mark each link as *verified on VirusTotal on YYYY-MM-DD*.

📌 **Style rule.** Never start a step with *"Click here"* or *"Press the obvious button"*. State **what** you are doing and **why** it works. The grader is looking for evidence that *you* understood, not that you transcribed a tutorial.

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# Side notes — alternative hypervisors for those who already have one

If you arrive in this module with a different hypervisor already running on your laptop, you do not have to install VirtualBox. Use whichever of the following you already trust:

- **VMware Workstation Pro** — free for personal use since May 2024. Snapshot performance is excellent and the GUI (Graphical User Interface) is more polished than VirtualBox.
- **KVM** (Kernel-based Virtual Machine) on a Linux host, driven by virt-manager — gives near-bare-metal performance, zero GUI tax, ideal on a Linux laptop.

- **[Proxmox VE]** (Virtual Environment) — turn an old desktop or NUC (Next Unit of Computing) into a dedicated home-lab hypervisor with a web GUI. Best option if you can spare a second machine.

If you switch hypervisor, your screenshots will look different from the classroom examples. Add a one-paragraph note at the top of your chapter explaining the swap and why you made it.

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## Alternative open-source tools — Day 05 Demo

A short list of equally legitimate paths to the same destination, in case you want to explore beyond the canonical recipe:

- **VMware Workstation Pro** — desktop hypervisor, free for personal use.
- **KVM** + **virt-manager** — native Linux hypervisor with a clean management GUI.
- **Proxmox VE** — turn-key hypervisor for a dedicated lab machine.
- **EVE-NG Community Edition** (Emulated Virtual Environment — Next Generation) — full enterprise topologies inside one Linux VM, popular alternative to GNS3.
- **Containerlab** — network topologies in Docker containers, near-instant boot, very lightweight on RAM.
- **Cisco DevNet Sandboxes** — free, always-on public sandboxes for Cisco gear; no install needed.
- **draw.io** — free diagramming editor for your topology figures.

Each link has been verified with VirusTotal at the time this pack was published. Re-verify before you click.

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## Day 05 Demo — Lab safety reminder (DO and DON'T)

- **DO** keep VirtualBox networking on host-only or NAT — **never bridged** to the campus or your home network unless explicitly instructed.
- **DO** take a snapshot before every change so you can roll back in seconds.
- **DO** click *End Lab* in AWS Academy when you finish — credits are real money and they evaporate overnight if you forget.
- **DO** change every default password before exposing any service, even inside the lab.
- **DON'T** install lab VMs on a personal device you also use for online banking, work email or school accounts.
- **DON'T** copy AWS root credentials into any public document — they appear once in the Learner Lab and cannot be rotated.
- **DON'T** scan or probe anything outside `10.10.10.0/24` from these VMs.

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# Day 06 — Step-by-step Hands-on Lab 3 (2 hours)

## Wireshark · SNMP · Wazuh · Splunk, with pfSense / OPNsense in the middle, on multiple operating systems

### Chapter 2 of your Course Procedure Manual — Lab 3 of 6

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## Why this Lab matters

You installed the *plumbing* in the Day 05 Demo. Today you install the *brain*: the SIEM-style (Security Information and Event Management) platform that lets you sit in front of one web dashboard and answer the question *"did anything bad happen this week?"*. By the end of this Lab you will:

- See **Wireshark** capturing live traffic mirrored from the OPNsense firewall.
- Watch **SNMP** polls travelling on the wire and being interpreted by your management station.
- Get a real **Wazuh** alert on a real Windows endpoint after a real (but harmless) triggering action.
- Forward that Wazuh alert into **Splunk Free** and find it again with one Splunk Processing Language (SPL) query.
- Review logs on **three different operating systems** so you start spotting where each OS hides its evidence.

That four-step chain is a simplified version of what runs inside every modern SOC (Security Operations Center).

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## Prerequisites — already completed in the Day 05 Demo

Tick every item before going further. There is no point in installing Wazuh on a broken lab.

- [ ] VirtualBox host-only network `10.10.10.0/24` operational.
- [ ] VMs running: `cyb01-win10` (10.10.10.20), `cyb01-ubuntu` (10.10.10.30), `cyb01-kali` (10.10.10.40), `cyb01-opnsense` (LAN 10.10.10.1, WAN via NAT).
- [ ] Internet egress works from each VM (test with `curl -I https://www.cyber.gc.ca/` from Ubuntu).
- [ ] All VMs snapshotted as `01-networked`.

If any item is unchecked, fix it first.

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## Phase 1 — Wireshark on the capture VM (15 min)

The **Ubuntu VM** becomes the dedicated capture host. Mirror the OPNsense LAN switch traffic to it. In GNS3: right-click the switch port → *Configure* → *Port Mirroring* → mirror to the Ubuntu's port.

Open the Ubuntu console (or SSH into it) and run:

```
# Install Wireshark + tshark, the headless command-line capture tool.
sudo apt update && sudo apt install -y wireshark tshark

# Add yourself to the wireshark group so non-root captures work.
sudo usermod -aG wireshark $USER
newgrp wireshark

# Start a 30-second capture on the LAN interface. The name is usually enp0s3
# inside a VirtualBox VM — confirm with `ip link` if unsure.
sudo tshark -i enp0s3 -a duration:30 -w /tmp/cyb01-baseline.pcapng
```

Generate traffic *from* the Windows VM while the capture is running: open three different websites, ping `8.8.8.8` half a dozen times, hit the OPNsense web GUI.

Copy `/tmp/cyb01-baseline.pcapng` to your host laptop (`scp`, shared folder, or copy-paste through the VM's clipboard sharing) and open it in Wireshark. Apply the display filter:

```
http or dns or icmp
```

**Screenshot — Figure 1:** *Wireshark window showing the filtered traffic, including at least one DNS query and one HTTP request, with the display filter visible in the toolbar.*

📺 **Video (≈ 10 min):** Search the [David Bombal channel] for *"Wireshark for beginners"*. Open the channel directly; do not download or rehost.

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## Phase 2 — SNMP on OPNsense and verification (15 min)

1. Log into the OPNsense web GUI at `https://10.10.10.1`.
2. *Services* → *SNMP* → *Enable*. Set:
   - Community string: `labdemo`
   - Location: `Cyber.SoHo Lab`
   - Contact: `cyber@cyber.soho.local`
3. *Save & Apply*.

From the Ubuntu VM:

```
# Install the Net-SNMP tools and the MIB (Management Information Base)
downloader.
sudo apt install -y snmp snmp-mibs-downloader

# Pre-populate the local MIB directory so the OIDs decode to readable names.
sudo download-mibs

# Walk the system tree to confirm SNMP is alive.
snmpwalk -v2c -c labdemo 10.10.10.1 1.3.6.1.2.1.1
```

Note the values returned for `sysDescr`, `sysName`, and `sysUpTime`. Capture them as Figure 2.

In parallel, run a Wireshark capture on the Ubuntu VM filtered on `snmp`. You will see the `labdemo` community string travelling **in cleartext**. That is the entire selling point of SNMPv3 — write that observation into your Reflection section.

**Screenshot — Figure 2:** *the full `snmpwalk` output with sysDescr, sysName and sysUpTime clearly visible.*

**Screenshot — Figure 3:** *Wireshark capture showing the same SNMP packets, with the community string `labdemo` highlighted in the packet bytes pane.*

⚠ **Reflection prompt.** In your chapter, answer: *if this were a production network, what version of SNMP would you use and why?* Cite RFC 1157 (SNMP v1 / v2c, no encryption) and RFC 3414 (SNMPv3 USM — User-based Security Model, with authentication and encryption).

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# Phase 3 — Install Wazuh single-node on the Ubuntu VM (30 min)

**Resource note.** Before starting, shut the Ubuntu VM down and bump its RAM allocation to **4 GB** in VirtualBox Manager. The Wazuh indexer (which is OpenSearch under the hood) will crash on the install otherwise.

```
# Sanity check: confirm we are on the Ubuntu lab VM, not a production host.
hostname

# Download the official Wazuh installation assistant from the project's CDN.
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh

# Run the assistant in all-in-one mode (-a): server + indexer + dashboard on
one box.
sudo bash ./wazuh-install.sh -a
```

The script ends by printing a username (`admin`) and a long temporary password. **Save them in your password manager immediately.** You will not be shown them again.

1. From your laptop browser open `https://10.10.10.30/`. Accept the self-signed certificate warning (in production you would replace this with a proper certificate; not in scope for this Lab).
2. Log in with the `admin` credentials.

**Screenshot — Figure 4:** *Wazuh login screen with the URL bar visible.*

**Screenshot — Figure 5:** *Wazuh overview dashboard with zero agents enrolled.*

📺 **Video (≈ 13 min):** Search the [Wazuh official channel]() for *"Wazuh quickstart"*. Use the official channel link; do not mirror or rehost.

## Enrol the Windows agent

1. In the Wazuh dashboard → *Add agent* → select **Windows MSI** (Microsoft Installer package).
2. The dashboard generates a customised one-line install command. Copy it.
3. On the Windows VM, open *PowerShell as Administrator*. Paste the command. It downloads the MSI from `packages.wazuh.com`, installs it, and registers the host with the server.
4. From an Administrator command prompt:

```
NET START WazuhSvc
```

5. Back in the Wazuh dashboard → *Agents*. The Windows VM should appear *Active* within roughly 30 seconds.

**Screenshot — Figure 6:** *Wazuh agents list with the Windows VM showing Active status.*

## Generate a real alert (the fun part)

On the Windows VM, open *PowerShell as Administrator* and run:

```
# Deliberately fail five logons in a row -- Wazuh's rule set will alert on
this.
1..5 | ForEach-Object {
    Start-Process -FilePath "runas.exe" `
                  -ArgumentList "/user:nonsense$_ cmd.exe" `
                  -ErrorAction SilentlyContinue
}
```

Wait about thirty seconds. Open the Wazuh dashboard → *Security events* → *Failed logins*. You will see the alerts arrive.

**Screenshot — Figure 7:** *Wazuh dashboard showing the five failed-logon alerts with timestamps and source user names.*

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# Phase 4 — Install Splunk Free on the same VM (25 min)

Splunk Free can coexist with Wazuh on the same 4 GB Ubuntu VM for this Lab. In production you would put them on separate boxes — note that in your Reflection.

```
# Register for a free account on splunk.com, then download the .deb manually.
# (Splunk requires the cookie-protected click-through, so wget alone will not
work.)
# Move the downloaded file to the Ubuntu VM and place it in the home
directory.

# Verify SHA-256 against the hash shown on the Splunk download page.
sha256sum splunk-9.x.x-amd64.deb

# Install.
sudo dpkg -i splunk-9.x.x-amd64.deb

# Start once to accept the licence and set the admin password.
sudo /opt/splunk/bin/splunk start --accept-license

# Enable auto-start so it survives a reboot.
sudo /opt/splunk/bin/splunk enable boot-start
```

1. Browse to `http://10.10.10.30:8000/`. Log in with the admin credentials you set during install.
2. *Settings → Licensing → Change license group → **Free***. Restart Splunk when prompted.
3. *Settings → Data Inputs → HTTP Event Collector → Global Settings → Enable*.
4. *Settings → Data Inputs → HTTP Event Collector → New Token → Name =* `wazuh-feed`. Save and copy the token.

**Screenshot — Figure 8:** *Splunk Web showing the Free licence active under Settings → Licensing.*

**Screenshot — Figure 9:** *the HEC (HTTP Event Collector) token you just created. **Mask all digits after the first eight characters** with a black rectangle before submitting your PDF — the token is a credential.*

## Configure Wazuh to forward alerts to Splunk

Edit `/var/ossec/etc/ossec.conf` on the Wazuh server and add the following integration block. Replace `<TOKEN>` with the value you just saved.

```
<integration>
  <name>custom-splunk</name>
```

```
  <hook_url>https://127.0.0.1:8088/services/collector</hook_url>
  <api_key><TOKEN></api_key>
  <level>7</level>
  <alert_format>json</alert_format>
</integration>
```

Then restart the Wazuh manager:

```
sudo systemctl restart wazuh-manager
```

Trigger a couple more failed logons on the Windows VM. In Splunk Web, run:

```
index=main sourcetype="wazuh:alerts" | head 20
```

**Screenshot — Figure 10:** *Splunk search results showing the Wazuh JSON alerts you just forwarded, with the search bar visible.*

📺 **Video (≈ 12 min):** Search the [Splunk YouTube channel]() for *"Splunk Wazuh integration"*, or read the [official Wazuh — Splunk integration guide](). Use the channel link directly.

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# Phase 5 — Configure alerts and explore logs across operating systems (15 min)

The same security event lives in different places on every OS. Knowing where to look is half the job.

## On Linux (Ubuntu — the Wazuh manager)

Tail the manager's live alert decisions:

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Trigger one more failed logon on the Windows VM. Watch the alert appear live.

**Screenshot — Figure 11:** *terminal showing the live `alerts.log` with the new failed-logon entry highlighted.*

## On Windows (the agent host)

Open the Wazuh agent log:

```
C:\Program Files (x86)\ossec-agent\ossec.log
```

Open it in Notepad++ or Visual Studio Code. Find the line where the agent confirms it received the alert from the manager.

**Screenshot — Figure 12:** *Windows agent log open in your editor, with the receipt-confirmation line highlighted.*

### On OPNsense (the firewall)

1. *System → Settings → Logging / Targets → Add Target*.
2. Remote IP: `10.10.10.30`, Port: `514`, Transport: `UDP`. Save & apply.

Confirm syslog is landing on the Ubuntu VM:

```
sudo tcpdump -i any -n udp port 514
```

Generate some firewall activity by browsing from the Windows VM. You should see syslog packets arrive in real time.

**Screenshot — Figure 13:** *tcpdump output showing live syslog UDP packets from 10.10.10.1 to 10.10.10.30, with at least one full packet visible.*

Three operating systems, three different log locations, one central dashboard. This is exactly the architectural value of a SIEM (Security Information and Event Management) platform — you stop hunting for evidence and start asking questions.

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## Phase 6 — Wrap up and snapshot (5 min)

For every VM that changed during this Lab, take a fresh snapshot named `02-monitoring-stack`. This is your roll-back point for Lab 4. Verify the snapshots exist in *VirtualBox Manager → Snapshots* before you shut anything down.

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## Phase 7 — Write your manual chapter (homework, ≈ 90 min)

Use the same Course Procedure Manual chapter template as Chapter 1. Mandatory sections:

1. **Title page.**
2. **Purpose** — one short paragraph in your own words.
3. **Topology diagram** — redraw, showing Wazuh + Splunk + agents on the network you built in Chapter 1.

4. **Procedure** — Phases 1 to 6 above, every figure captioned with at least one sentence of context.
5. **Evidence** — at least one search you wrote in Splunk + one rule trigger in Wazuh + one Wireshark display filter + one `snmpwalk` output. Each becomes a numbered figure.
6. **Troubleshooting** — three or more real issues you hit (we promise you will hit at least three). Examples to inspire you: *"Wazuh install died on the indexer — fixed by raising VM RAM to 4 GB and re-running with the `--ignore-check` flag"*; *"Splunk HEC returned 403 — fixed after noticing I had pasted a leading space into the token field"*.
7. **Reflection** — 200 words minimum, in your voice, answering at least:
    - Why is sending alerts to *both* Wazuh and Splunk useful (or wasteful, depending on the shop)?
    - At what daily log volume would you split Wazuh and Splunk onto separate VMs?
    - Which OS surprised you most when you looked at where it stores its security events?
    - What was harder than you expected?
8. **Sources** — every link, every video, every documentation page you opened, each marked *verified on VirusTotal on YYYY-MM-DD*.

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# Alternative open-source tools — Day 06 Lab 3

If your lab box cannot handle Splunk + Wazuh together, or if you simply want to broaden your portfolio, swap in any of these:

- **OpenSearch** + **OpenSearch Dashboards** — open fork of Elasticsearch + Kibana, replaces Splunk almost feature-for-feature. Wazuh actually uses OpenSearch internally, so a Wazuh + OpenSearch combination is a single stack rather than two.
- **Graylog Open** — lighter ingest pipeline than Splunk or OpenSearch, very strong on stream-based routing rules.
- **Security Onion 2** — one ISO that bundles Wazuh + Zeek + Suricata + Elastic + CyberChef; zero integration work needed.
- **TheHive** + **Cortex** — open-source case-management and observable-analysis platform, perfect downstream of Wazuh.
- **Snort 3** — the original signature-based IDS (Intrusion Detection System), still maintained by Cisco Talos, alternative to Suricata.
- **Zeek** (formerly *Bro*) — network-security-monitoring framework that produces connection logs rather than signature alerts; complementary to Suricata and Snort.
- **Suricata** by the Open Information Security Foundation — multi-threaded IDS / IPS (Intrusion Prevention System), feeds JSON alerts straight into Wazuh.
- **Fluent Bit** — modern, lightweight log shipper, much smaller footprint than Logstash.
- **Vector** — open-source log and metric router with one unified configuration language; handy when you have many sources and one destination.
- **Net-SNMP suite** — the canonical implementation of `snmpwalk`, `snmpget`, `snmptrap`, `snmpd`. Already used in Phase 2.

- **pfSense** — the other major free FreeBSD-based firewall. Most of this Lab works identically; the menu paths differ.

Each link has been verified on VirusTotal at publication time. Re-verify before clicking — feeds can change between publication and your reading.

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## Day 06 Lab 3 — Lab safety reminder (DO and DON'T)

- **DO** isolate the lab on a host-only network only.
- **DO** rotate the Wazuh `admin` password and the Splunk `admin` password **before any screenshot you put in your manual**. Show the change of password as a procedural step; never the password itself.
- **DO** verify every download URL with VirusTotal before clicking it, even on official vendor pages.
- **DO** snapshot before you change anything you cannot easily undo.
- **DON'T** publish your lab `.pcap` files, your Splunk HEC token, or your Wazuh API key to GitHub, Discord, Pastebin, or any cloud drive. Treat them as production secrets in training — that habit will save your career one day.
- **DON'T** run any of the commands in this Lab against a system you do not own or do not have written authorisation to test. The legal exposure under Canadian Criminal Code s.342.1, US CFAA 18 U.S.C. § 1030, EU Directive 2013/40/EU, and Spain's Ley Orgánica 10/1995 is real, and ignorance is not a defence.
- **DON'T** install the Wazuh agent or Wireshark on a personal device you also use for online banking, work, or school accounts.

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# Useful links — quick reference

A deduplicated index of every link used above, grouped by purpose. Verify each one with the VirusTotal URL scanner before clicking from a corporate machine.

### Hypervisors and lab tools

- VirtualBox and its downloads page.
- GNS3 and its download page.
- VMware Workstation Pro, free for personal use.
- KVM + virt-manager.
- Proxmox VE.
- EVE-NG, Containerlab, Cisco DevNet Sandboxes.
- draw.io — diagram editor.

### Operating-system images

- Microsoft Evaluation Center — legal Windows 10 / 11 trial ISOs.
- Ubuntu Server downloads.
- Kali Linux.
- OPNsense · pfSense.

### Cloud

- AWS Academy login · AWS Free Tier.
- AWS CloudWatch · GuardDuty.

### Monitoring stack covered in Lab 3

- Wireshark + tshark man page.
- Net-SNMP suite.
- OPNsense SNMP docs · pfSense SNMP docs.
- Wazuh documentation · Windows agent install guide · Wazuh — Splunk integration guide.
- Splunk Enterprise download · Splunk Free reference.
- Sysmon.

### Open-source alternatives

- OpenSearch, Elasticsearch, Kibana, Filebeat, Logstash.
- Graylog, Security Onion, TheHive, Cortex.
- Snort, Zeek, Suricata, OISF (Open Information Security Foundation).
- Fluent Bit, Vector.
- CyberChef.
- Cisco Talos.

### Standards and protocols

- IETF (Internet Engineering Task Force).
- RFC 1157 — SNMP v1.
- RFC 3414 — SNMPv3 USM.
- RFC 5424 — syslog.

### Submissions and procedures

- OmniVox student portal — your LEA DropBox lives here.
- VirusTotal URL scanner — verify every download URL.

### Threat intelligence and reference bodies

- MITRE ATT&CK — adversary tactics, techniques and procedures.
- VirusTotal · AbuseIPDB.
- CISA (US Cybersecurity and Infrastructure Security Agency).

- Canadian Centre for Cyber Security.
- OSFI (Office of the Superintendent of Financial Institutions, Canada) · OSFI Guideline B-13.
- ACM Code of Ethics · IEEE Code of Ethics.

**Legal references (four jurisdictions)**

- Canadian Criminal Code s.342.1 — Unauthorized use of computer.
- United States CFAA, 18 U.S.C. § 1030.
- EU Directive 2013/40/EU on attacks against information systems.
- Spain — Ley Orgánica 10/1995 del Código Penal.

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Before downloading any installer, opening any document, or running any script reached via a link in this Pack — **even from an official vendor site** — paste the destination URL into VirusTotal URL scanner and verify the SHA-256 hash of any downloaded file against the vendor's signed checksum. Treat every download as untrusted until you have personally verified it. The student is responsible for this verification; Cyber.SoHo accepts no liability for any link that becomes malicious after publication.

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# Liability and ethics — the tools in this Pack can be misused

The tools covered (Wireshark, Nmap, SNMP utilities, Wazuh, Splunk, OPNsense, pfSense, Suricata, Zeek, and the rest) are powerful **dual-use** technologies. They can defend a network and they can attack one; the legal distinction lives entirely in **authorisation**. Students agree to use them:

- **only inside the isolated lab environments described in this Pack**, on virtual machines they own;
- **only against systems they own or have explicit, written, dated authorisation to test**;
- **never** against the campus network, residence Wi-Fi, employer network, internet-service-provider network, neighbour's network, or any other system they do not unambiguously own or control.

Unauthorised scanning, packet sniffing, intrusion, lateral movement, credential capture, or data exfiltration is a criminal offence under, among others, the **Canadian Criminal Code s.342.1**, the **United States Computer Fraud and Abuse Act, 18 U.S.C. § 1030**, **EU Directive 2013/40/EU** on attacks against information systems, and **Spain's Ley Orgánica 10/1995 del Código Penal**. Conviction can carry imprisonment, fines, deportation, and lifetime employment exclusion from the cybersecurity profession. **Ignorance of the law is not a defence.**

Students are also expected to adhere to the **ACM Code of Ethics and Professional Conduct** and the **IEEE Code of Ethics** — the two foundational professional codes for our field. Read them once; revisit them when in doubt.

# Lab safety — what to DO and what NOT to DO

## DO

- Keep the entire lab on an **isolated host-only or NAT virtual network**. Never bridge to the campus, residence, or home network unless the instructor explicitly told you to.
- Take a **snapshot of every VM before any change** you cannot easily undo.
- **Change every default password** before exposing any service, even inside the host-only lab.
- **Verify every URL** you click via VirusTotal.
- **End the AWS Academy Lab** session as soon as you finish — credits drain even when you are not actively using them.
- **Document every troubleshooting incident** as you go — the freshest memory makes the cleanest manual entry.

## DON'T

- **Don't scan, probe, sniff, or attack** the institutional network, your residence Wi-Fi, your ISP, your employer's network, or any third-party address you do not personally own or have written authorisation to test.
- **Don't install** the Wazuh agent, Wireshark, Nmap, or any other Lab tool on a personal device you also use for online banking, employer work, school accounts, or personal communications.
- **Don't post** Wazuh admin credentials, Splunk HEC tokens, captured `.pcap` files containing real personal data, or any other lab artefact to GitHub, Discord, Pastebin,

Slack, public cloud storage, or any other publicly searchable medium. From the day you start in this profession, these strings are part of *the secrets you must protect*.
- **Don't copy** AWS Learner Lab root credentials into any document — they cannot be rotated in the Academy environment.
- **Don't reuse passwords** between the lab and any real-life account you own.
- **Don't rely** on the screenshots in this Pack as evidence that you ran the steps — produce your own.

# Author indemnity — clear and unambiguous

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